Stress-Dependent Cardinal Boundaries of Native Assignment States in Aerial Tiny-Object Detection

Yede Liu[a], Fang Wang[a] and Jun Li[a,*]

[a] *College of Physics and Electronic Engineering, Sichuan Normal University, Chenglong Campus, No. 1819, Section 2, Chenglong Avenue, Longquanyi District, Chengdu, 610101, Sichuan, China*

ARTICLE INFO

Keywords:
tiny-object detection
aerial imagery
one-to-one assignment
geometric stress testing
cardinal boundaries
assignment-state measurement

ABSTRACT

Many end-to-end dense detectors retain one post-conflict assigned candidate for one-to-one (O2O) supervision, but the stability of this discrete state under controlled geometric stress is poorly characterised. We developed a read-only audit that moves only a focal ground-truth centre while freezing model outputs, parameters, candidate grids, object extent, and other ground truths. Implementation qualification preceded interpretation through native-path checks and exact full-grid replay. On AI-TOD-v2, fixed one-pixel replay yielded an O2O 8–16-minus-16–32 px fragility contrast of 14.37 percentage points (pp), whereas equivalent-area-side replay yielded an O2O contrast of 0.13 pp and an assigned-set Top-1 contrast for one-to-many (O2M) supervision of -17.45 pp. The apparent scale response therefore depended on the stress contract rather than indicating a single intrinsic O2O penalty. Eligibility locking and pathway analysis separated hard eligibility changes from within-set rank exchange. Full-domain $1/1024$ replay yielded restricted mean cardinal boundary distances of 0.120 for O2O and 0.100 for the assigned-set Top-1 state through $\tau = 0.125$; Rank–Set and positive-count analyses further distinguished Top-1 turnover from positive-set persistence. Assigned-relative margin aligned with observed rank-crossing distance but supplied no stable incremental diagnostic value. Primary mechanistic estimates are seed-0 AI-TOD-v2 discovery results; cross-dataset, stage, detector-contract, and training-seed analyses are sensitivity or replication. The audit resolves assignment change into stress-dependent eligibility, rank, set, and cardinal-boundary states without treating the branch contrast as a treatment effect or calibrated failure probability.

1. Introduction

Aerial tiny-object detection places discrete training decisions under unusually coarse geometric resolution. Dense candidates lie on stride-dependent lattices, while many NMS-free detectors use a one-to-one (O2O) branch that retains a single post-conflict assigned candidate for each ground-truth object [6, 7, 9, 10]. Even a one-pixel centre displacement can change the frozen assignment winner. A small displacement can therefore alter not only overlap or score, but also the identity, rank, eligibility, or positive-set membership used to construct supervision. These are properties of the native assignment state rather than postprocessed detections.

Existing tiny-object methods modify localisation similarity, receptive-field-aware assignment, scale-aware labels, feature resolution, or supervision density [1, 2, 3, 4, 14, 15, 16]. Dense-detector assignment methods likewise change candidate selection through adaptive matching, optimal transport, task alignment, or differentiable assignment [17, 18, 19, 20, 21, 22]. Such methods address how positives should be constructed. They do not, by themselves, measure how stable the resolved native assignment state is after training, nor whether an apparent scale response depends on the geometric stress used to probe it.

*Corresponding author: Jun Li (lijun2022@sicnu.edu.cn).

liuyede0718@gmail.com (Y. Liu); fang66320@gmail.com (F. Wang); lijun2022@sicnu.edu.cn (J. Li)

ORCID(s): 0009-0004-1760-860X (Y. Liu)

The fact that one input pixel represents a larger fractional displacement for a smaller object is elementary. The unresolved questions are whether an apparent assignment-state scale effect survives an equivalent-area-side stress, which native states change, and whether a Top-1 exchange also implies turnover of the positive supervision set.

Measuring this state is non-trivial for three reasons. First, the instrument must reproduce native eligibility, Top-k selection, conflict resolution, tie handling, and numerical precision closely enough that a reported identity change is not a tracing artefact. Second, O2O identity, O2M rank, and O2M assigned-positive-set membership are different states; for the diagnostic questions considered here, a single endpoint candidate-change variable is insufficient to distinguish the audited native states. Third, base-plus-four-direction replay observes responses only at one displacement magnitude, but does not show how close the first boundary lies, whether a trajectory is right-censored, or which native stage produced the first divergence.

We therefore develop a qualified read-only measurement of the frozen native assigner. The focal ground-truth centre is replayed under a fixed-pixel or equivalent-area-side displacement while image pixels, extracted features, model parameters, candidate grids, box size, and all non-focal ground truths remain fixed. The measurement contract is explicit: a one-pixel shift asks about sensitivity to an absolute input displacement, whereas equivalent-area-side replay scales displacement by $\sqrt{wh}$ and is not the same axis-specific fraction of width and height for a rectangular object. Neither contract is treated as a correction for the other. Analytical oracles, property invariants, mechanism mutations, stage-wise native/reference parity, exact full-grid focal-row/native equivalence, and an explicit grid-resolution gate qualify the implementation before scientific interpretation. The audit then combines endpoint replay, eligibility locking, first-divergence and multi-stage anatomy, O2O pre-resolution controls, O2M rank–set decomposition, and cardinal-direction boundary analysis. Weighted Kaplan–Meier and Aalen–Johansen summaries retain censoring and competing boundary causes.

The contributions are threefold:

- **Stress-contract dependence.** On AI-TOD-v2, fixed-pixel and equivalent-area-side replay led to qualitatively different scale conclusions. Sensitivity analyses across four protocol-eligible seeds, VisDrone, archived stages, and a compatible detector contract showed that the branch-wise response structure was not confined to a single checkpoint or analysis setting.
- **State-specific assignment anatomy.** Eligibility locking, first-divergence analysis, and O2M Rank–Set decomposition separate eligibility transitions, within-set rank exchange, Top-1 turnover, and positive-set turnover. For the diagnostic questions considered here, a single endpoint candidate-change variable is insufficient to distinguish the audited native states.
- **Cardinal boundary persistence.** Exhaustive $1/1024$ replay along four cardinal rays quantifies first-departure distance, censoring, competing boundary causes, and restricted mean persistence, revealing different observed persistence structures for assigned O2O identity and O2M assigned-set Top-1.

Implementation qualification is reported separately as a prerequisite for interpreting these measurements.

The practical role of the audit is not to prescribe a replacement assigner, but to determine whether an apparent assignment effect reflects the stress coordinate, a change in candidate ranking, a change in the supervision set, or a shift in the observed assignment boundary. Accordingly, a fixed-pixel scale effect can reflect the chosen stress coordinate rather than an intrinsic scale law; O2M Top-1 turnover need not imply positive-set turnover; and endpoint flip/no-flip omits distance, censoring, cause, and return information.

2. Related work

2.1. Tiny-object geometry and scale-aware assignment

NWD, RFLA, SimD, dynamic coarse-to-fine learning, and normalized Gaussian label assignment reduce the brittleness of overlap-based matching for tiny objects [1, 2, 3, 4, 14]. Recent scale-aware assigners use virtual-anchor calibration or normalized Bhattacharyya distance [15, 16]; UGS addresses localisation gradients, and DALA separates dense and sparse objects by spatial density [5, 13]. These methods modify geometry, optimisation, positive construction, or supervision density.

More broadly, dense-detector assignment includes adaptive and probabilistic candidate selection [17, 18], optimal transport and task alignment [19, 20], and differentiable matching [21, 22]. These approaches optimise positive construction.

2.2. End-to-end dense detection and dual supervision

DeFCN introduced prediction-aware O2O assignment with an O2M auxiliary objective [6]. OneNet used class-aware matching for non-redundant prediction [7]. EOOD combined an oriented O2O assigner, gap regularisation, and O2M auxiliary heads [23]; DATE restored O2M supervision to accelerate O2O convergence [8]. Hybrid methods add grouped or auxiliary supervision while retaining one-to-one inference [24, 25, 26], and PaQ-DETR couples dynamic queries with quality-aware O2M assignment [27]. YOLOv10 aligned O2M and O2O criteria [9], whereas YOLO26 combined dual heads with direct LTRB regression, Progressive Loss, and small-target-aware positive coverage [10].

2.3. Different meanings of assignment ambiguity

Stable-DINO studied inter-layer Hungarian-matching instability, and many-to-one pedestrian matching challenged arbitrary selection among similar proposals [28, 29]. Both changed training to improve robustness. TCD linked assignment ambiguity to inconsistent preset intermediates and model predictions [11]. Its dynamic sample set addresses conventional single-stage assignment. DS-Det located query ambiguity in shared-weight decoders trained with O2M and O2O objectives [12]. O2F assigned dynamic soft supervision to several ambiguous anchors [30]. Output-level diagnostic tools such as TIDE decompose detection errors without accessing the training assignment path [31]. Our estimand retains one native O2O winner and measures its local stability. Existing assignment studies mainly replace matching or supervision rules. Our contribution is a post-training measurement of the native post-conflict assigned identity, not a replacement assigner.

These strands address different objects of analysis: training-time matching-stability work studies or modifies matching during optimisation; output-level diagnosis decomposes errors after predictions are produced; and image/metamorphic robustness testing perturbs inputs and measures changes in model outputs [31, 34, 35]. This work instead performs post-training, frozen-output, focal-GT-only replay of the native assignment path. It therefore makes no first-ever claim about assignment stability or robustness-boundary analysis; its scope is the stated diagnostic measurement contract.

2.4. Post-training neural-system testing and diagnostics

DNN testing studies how to select informative test inputs, estimate fault-revealing behaviour, and reduce labelling cost [34, 35]. Other diagnostic work uses near-miss concepts for model comparison and debugging [36], or treats anomalous behaviour as dependent on the evolving training state [37]. These studies analyse prediction behaviour, representations, test selection, or training anomalies. Our audit instead targets the resolved native assignment state used to construct dense-detector supervision and measures its response while network outputs remain frozen.

3. Methods

3.1. Study design and frozen detector contracts

The study is a post-training diagnostic analysis of native assignment state. The principal contract was a YOLO26s detector with native O2M and O2O task-aligned assignment [10]. A compatible YOLOv10-S contract provided a second-detector replication [9]. AI-TOD-v2 was the principal dataset and VisDrone was used for cross-dataset sensitivity [32, 33]. The VisDrone sensitivity used a separate VisDrone-trained YOLO26s seed-0 checkpoint, not AI-TOD-trained weights replayed on VisDrone. All replay analyses used frozen checkpoints. No replay result was used to update weights, alter inference, or select a replacement checkpoint.

The discovery sample comprised 300 outcome-blind stratified AI-TOD-v2 validation images. The primary scale domain was prespecified as $8 \leq \sqrt{w_g h_g} < 32$ input pixels and divided at 16 px, using the same scale-defined image strata for AI-TOD-v2 and VisDrone. Objects below 8 px were outside this estimand and were not retrospectively introduced as a new outcome stratum. Dataset-specific sampling weights were the inverse image-inclusion probabilities. Unless stated otherwise, uncertainty was estimated with 5,000 bootstrap replicates that resampled image clusters within the frozen sampling strata. These intervals quantify image-sampling uncertainty conditional on the audited seed-0 checkpoint; they do not include training-seed or checkpoint-selection variability. No family-wise multiplicity correction was applied across the discovery and sensitivity matrix; its intervals are descriptive uncertainty summaries rather than confirmatory family-wise tests. The statistical unit is stated for every estimand because object-level fragility, directional transitions, candidate ranks, positive sets, and image-pair spillover are not interchangeable.

Training-seed sensitivity used every released artifact satisfying the frozen common contract: `weights/yolo26s.pt` initialisation, identical data and training arguments, and embedded zero-based epoch 280. Seeds 0–3 were eligible; seed 4 was excluded before outcome analysis because its recorded model argument points to a prior run's `last.pt`. Fixed and equivalent-area-side endpoint analyses retained strict cross-seed common-valid intersections of 6,318 and 6,317 GTs, respectively; the equivalent-area analysis also evaluated Rank–Set metrics on 25,165 legal directions per seed. Each of 20,000 crossed bootstrap replicates resampled four seeds with replacement and synchronously resampled image IDs within the frozen strata. These analyses remain sensitivity estimates with only four top-level seeds; eligibility, pathway, and dense-boundary results remain conditional on seed 0.

3.2. Native O2O identity, O2M states, and margin

The detailed path below is the principal YOLO26s contract; the YOLOv10-S replication retained the same alignment and conflict rules but used its native O2O Top-1 rather than YOLO26s Top-7 preselection, as tabulated in the Supplementary Material. For ground-truth object g, candidate i, and class c_g, native task alignment used

$$q_{ig} = p_{i,c_g}^{0.5} \left[\max\{\mathrm{CIoU}(b_i, g), 0\}\right]^6, \tag{1}$$

Equation (1) defines the alignment score reevaluated at each replay state. Candidate class scores p_{i,c_g} and decoded boxes b_i remain frozen, while the clipped-CIoU term changes with the replayed GT geometry. Candidate eligibility followed the detector's native small-target-aware region. With feature strides $(8, 16, 32)$, each GT dimension strictly smaller than 8 input pixels was replaced by 16 pixels for the eligibility calculation; other dimensions were unchanged. A candidate was eligible only when its anchor centre lay strictly inside the resulting rectangle, with all four signed edge distances greater than 10^{-9}. Eligible candidates were ranked in native float32 by q_{ig}. The O2O path retained the native Top-7 set. If one anchor was claimed by multiple GTs, it was assigned to the GT with maximal clipped CIoU using native `argmax` ordering; a final native Top-1-by-q operation among surviving claims produced the post-conflict assigned O2O identity. The O2M path used Top-10, applied the same native multi-GT conflict operation, and retained the post-conflict positive set. Exact ties followed native PyTorch `topk`/`argmax` ordering and were not replaced by a secondary abstract tie rule. If the post-resolution foreground and target-index state contained exactly one candidate for g, its index was denoted i_g^{assigned}. This index records assignment through `fg_mask` and `target_gt_idx`; strictly positive classification and box/DFL weight additionally requires positive normalised `target_scores`. We therefore use "assigned O2O identity" rather than treating assignment alone as proof of nonzero loss weight.

When a post-conflict assigned O2O identity and a positive alternative were both available, the native runner-up $i_g^{(2)}$ was the highest-q positive alternative for the same object. The assigned-relative margin was

$$m_g = \frac{q_{i_g^{\mathrm{assigned}},g} - q_{i_g^{(2)},g}}{q_{i_g^{\mathrm{assigned}},g} + 10^{-12}}. \tag{2}$$

Equation (2) was evaluated only when both identities existed. Margin was undefined when the assigned O2O identity or a positive alternative was absent. Undefined competition was reported as a separate structural state and was never imputed as an extreme margin.

O2M was represented by several distinct states. The native pre-conflict identity is the largest-q candidate inside the native Top-10 set. The final assigned-positive set $\mathcal{P}_g$ contains candidates assigned to g after conflict resolution, and the assigned-set Top-1 identity is its largest-q element. Assigned-set Top-1 is a rank-state summary within the native positive set, not a uniquely supervised O2M sample;

every member of $\mathcal{P}_g$ remains part of the O2M supervision state. Rank state and set state were therefore analysed separately through Top-1 exchange, set equality, Jaccard similarity, and base-set retention.

3.3. Geometric stress and base-plus-four-direction replay

Only the focal ground-truth centre was moved. Image pixels, extracted features, parameters, candidate grids, decoded candidate boxes, focal width and height, and all non-focal ground truths remained fixed. For a base centre (x_g, y_g), fixed one-pixel replay used

$$\mathcal{D}_{\mathrm{px}} = \{(0,0), (\pm 1, 0), (0, \pm 1)\}. \quad (3)$$

Equivalent-area-side replay used $\kappa = 0.0625$ and

$$\delta_g = \kappa \sqrt{w_g h_g}, \qquad \mathcal{D}_{\mathrm{eq}} = \{(0,0), (\pm\delta_g, 0), (0, \pm\delta_g)\}. \quad (4)$$

Equations (3) and (4) define the two stress contracts. The name equivalent-area-side refers to the side length of a square with area $w_g h_g$; it does not imply equal directional fractions. For a rectangular GT, $\delta_g / w_g = \kappa\sqrt{h_g/w_g}$ and $\delta_g / h_g = \kappa\sqrt{w_g/h_g}$. Sensitivity analyses used $\kappa \in \{0.03125, 0.0625, 0.125\}$. Fixed and equivalent-area-side contracts answer different perturbation questions and were not pooled.

A secondary direction-normalised sensitivity used $\delta_x = \kappa w_g$ for left/right replay and $\delta_y = \kappa h_g$ for up/down replay at $\kappa = 0.0625$. The paired comparison used the identical 6,377-GT common-valid support. We defined aspect ratio as $\max(w_g/h_g, h_g/w_g)$ and stratified it as ≤ 1.5, $(1.5, 2.5]$, and > 2.5. Aspect-standardised rates used the pooled aspect-stratum distribution as a common target. Differences and contract contrasts used 5,000 paired stratified image-cluster bootstrap replicates.

For a native state $z_g(d)$ under displacement d, an object was stable when all legal shifted states equalled the base state and fragile when at least one legal shifted state differed. An unavailable base identity, missing assigned O2O identity, or absence of a valid relative competitor was retained as undefined according to the relevant estimand. Shifts that moved the GT outside the legal image domain were omitted rather than clipped. Directional analyses used each legal base-to-shift transition as the unit. Consequently, the object-level any-direction estimand has an opportunity-count dependence; a four-valid-direction point sensitivity is reported in the Supplementary Material.

The downstream diagnostic is a coverage-based miss. We previously labelled it FN@IoU50. It indicates that no frozen same-class prediction among the prespecified confidence and top-count outputs overlaps the focal GT by at least 0.5. Predictions are not consumed one-to-one, so this outcome is not the standard evaluator false negative.

Figure 1 summarises the resulting measurement object: a native post-conflict assigned identity is read from the unmodified assignment path, the focal GT alone is replayed through controlled centre shifts, and identity, eligibility, pathway, and continuous-boundary readouts are recorded without changing inference or supervision.

3.4. Instrument qualification and numerical contract

Instrument qualification preceded scientific interpretation. Six analytical oracle cases covered within-set rank reversal, eligibility transition, Top-7 membership transition, conflict reassignment, clipped-CIoU handling, and the historical taxonomy label "active disappearance". Seven metamorphic/property invariants checked candidate permutation, GT ordering, direction-processing order, score/CIoU ranking behaviour, conflict-off behaviour, and eligibility locking. Six prespecified mutations targeted CIoU clipping, Top-k cardinality, target-GT ownership, small-target eligibility expansion, missing-runner handling, and conflict resolution.

Native execution, the production tracer, and an independent NumPy reference were compared stage by stage. The broader deterministic parity set contained 300 base states and 5,251 directional states. A target-score qualification on the same set found zero assigned identities with nonpositive normalised target score among 8,079 base and 5,159 shifted assigned GT states. A separate executable edge case confirmed that `fg_mask` assignment alone does not guarantee positive weight, which motivates the assigned-identity terminology above. Continuous-boundary validation used a deterministic 30-image subset, one network forward per image, and an independent NumPy scanner that did not import production taxonomy helpers. These checks qualified replay-state calculations, but they did not constitute an end-to-end dry run of the sealed analyzer on a selected image with zero raw GT rows or on fixed-only common-support keys. That missing test coverage later invalidated the preregistered analyzer invocation.

Native float32 execution was normative. Candidate ordering, exact ties, eligibility comparisons, clipped-CIoU zero handling, margin epsilon, and boundary brackets followed the frozen numerical contract. Numerical differences inside a prespecified tolerance were labelled boundary cases rather than silently assigned to a scientific pathway.

3.5. Eligibility and first-divergence analyses

The eligibility-lock counterfactual held the base eligibility state fixed during replay and recomputed the remaining native path. It estimates how much of the observed response depends on crossing hard eligibility boundaries under that counterfactual; it is not a training intervention. The complementary eligibility-stable analysis conditioned on objects or directions whose relevant eligibility state remained comparable in both replay states. Because this conditioning is observed after replay, it changes the estimand and may induce selection.

Every directional O2O identity exchange received one mutually exclusive first-divergence label under the instrumented audit hierarchy: eligibility boundary, Top-7 membership transition, conflict reassignment, within-set geometry rank reversal, the historical taxonomy label "active

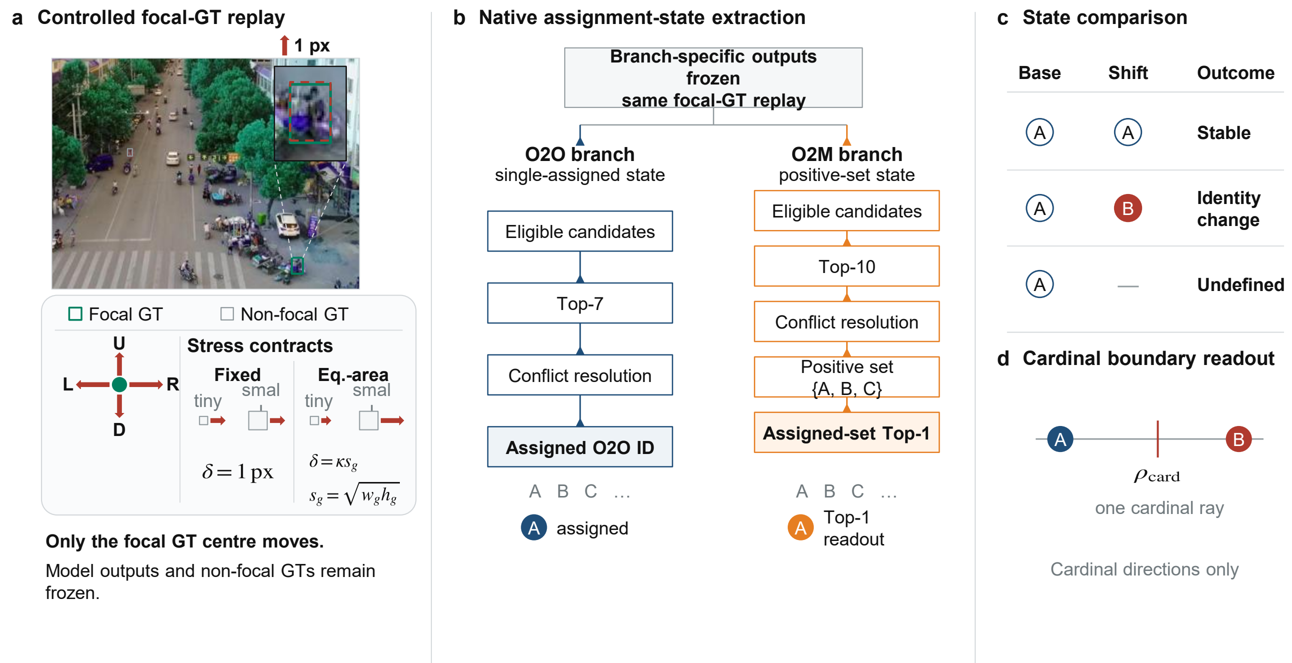

Figure 1: The read-only audit moves only the focal GT centre while every model output and non-focal GT remains fixed. (A) A registered AI-TOD-v2 crop and magnified inset show the exact base GT and its one-input-pixel upward shift. Fixed replay applies the same absolute displacement across object sizes, whereas equivalent-area-side replay scales displacement as $\delta = \kappa\sqrt{w_g h_g}$. (B) Branch-specific learned predictions are frozen separately under the same focal-GT replay contract; the panel does not imply shared scores or decoded boxes. Both native paths include conflict resolution. O2O terminates in one post-conflict assigned identity; O2M terminates in an assigned-positive set and its Top-1 rank readout. (C) Base-to-shift comparison defines stable, identity-change, and undefined states. (D) The first observed boundary is read along one cardinal ray and is not a two-dimensional minimum radius.

disappearance", or compound/other. The label is the earliest recorded native state difference, not a causal attribution. Clipped-CIoU-to-zero events were retained separately as overlapping geometry-clipping diagnostics.

For the pre-resolution control, the native pre-conflict Top-7 q winner was compared with the final post-conflict assigned O2O identity under fixed and equivalent-area-side replay. Scale-gap and object-fragility disagreements were computed within the O2O branch. Disagreements were then localised to the instrumented conflict stage.

3.6. O2M rank–set decomposition

For the base and shifted assigned-positive sets, exact equality, Jaccard similarity, and retention of the base set were recorded. Jaccard was defined only when the union was non-empty; empty/empty remained undefined rather than being assigned one. Base retention was undefined when the base set was empty. Empty-to-nonempty and nonempty-to-empty transitions were reported separately.

Assigned-set Top-1 changes were decomposed into rank-only exchange, set turnover with rank exchange, and set turnover without rank exchange. No post hoc Jaccard threshold was used. Fixed and equivalent-area-side contracts used the same image clusters, inverse inclusion weights, and 5,000 stratified image-cluster bootstrap replicates.

To assess mechanical positive-cardinality composition, a count-conditioned sensitivity analysis used the common-valid equivalent-area-side support. Exact-count standardisation used the pooled sampling-weighted distribution of native O2M positive count. The adequate-overlap analysis retained counts $K = 3, \ldots, 7$, each with at least 20 raw GTs in both size groups; coverage was reported for each group. This is a descriptive count-conditioned estimand rather than a fixed-K native state or causal size effect. Intervals used 5,000 stratified image-cluster bootstrap replicates, matching the v5 artifact.

3.7. Continuous cardinal-direction boundary persistence

Equivalent-area-side distance r was evaluated from zero to $r_{\max} = 0.25$ in four cardinal directions on the complete $1/1024$ grid. Every legal grid point was replayed in increasing radius order; the first sampled base-to-changed transition defined the observed boundary. The trajectory was not assumed monotone, so later returns such as $A \rightarrow B \rightarrow A$ were retained. The full-domain schedule prevents an event visible at the declared $1/1024$ resolution from being skipped, but it cannot resolve a transient interval narrower than one grid step. No GT clipping was allowed. Trajectories whose base box lay outside the legal image domain were excluded, and a trajectory with no event was censored at the smaller of $r_{\max}$ and its exact image-boundary legal radius.

For state type k, the first observed cardinal-direction boundary distance was ρ_{gdk}. It is not the minimum distance over the two-dimensional perturbation plane. Trajectories with no observed boundary through $r_{\max}$ were right-censored. Because the legal radius at which a focal GT exits the image can depend on image position, this geometric censoring is not a common administrative horizon across trajectories. Eligibility, Top-k, conflict, within-set rank, and other causes were treated as competing first boundaries where applicable. Weighted Kaplan–Meier survival $S_k(r)$ summarised overall state persistence, and Aalen–Johansen cumulative incidence summarised boundary causes. KM/AJ and the restricted mean cardinal boundary distance are descriptive summaries under the recorded administrative and geometric censoring rules; they are not interpreted as censoring-independent population persistence. Raw first-event incidence, cause counts, and return-transition reporting are preserved separately rather than replaced by these summaries. The restricted mean cardinal boundary distance was

$$\mathrm{RMCBD}_k(\tau) = \int_0^{\tau} S_k(r)\,\mathrm{d}r, \qquad \tau = 0.125. \tag{5}$$

Equation (5) was evaluated without a proportional-hazards assumption. The prespecified primary truncation, $\tau = 0.125$, is one eighth of the equivalent-area side, corresponding to approximately 1–4 px for objects with an 8–32 px equivalent-area side. Additional truncation-point sensitivities used the same frozen directional events; the paired O2O-minus-O2M contrast retained its direction at all six evaluated values of τ.

For fixed base assigned O2O candidate A and runner-up B, ρ_R was the first observed cardinal-direction distance at which their q order crossed while both remained comparable. An earlier eligibility, pre-Top-k, or conflict change was a competing censor; no crossing by $r_{\max}$ was a right censor. Margin–ρ_R association was reported only among observed crossings and was compared with the association between margin and observed eligibility-boundary distance. It is not a censoring-adjusted population correlation and does not establish population-level construct validity.

3.8. Spillover and secondary diagnostic analyses

Assignment spillover measured whether moving focal GT g changed the post-conflict assigned O2O identity of a non-focal GT h. The primary structural factor was whether g and h shared at least one native pre-conflict candidate claim in the base or shifted state. Pair-level rates used inverse image-inclusion weights and 5,000 stratified image-cluster bootstrap replicates. The full 300-image analysis was run only after a prespecified 30-image pilot met the gated scientific-value criterion.

Margin-shape sensitivity compared base covariates plus a linear margin term with the same covariates plus a restricted cubic spline margin basis. Five image-grouped out-of-fold splits were used. Spline knots were estimated only from each training fold and then applied unchanged to held-out image groups. ROC-AUC, PR-AUC, and log loss were reported for identity fragility and coverage-based miss. This analysis tests shape sensitivity, not whether margin is a causal or optimal predictor.

The outcome-blind budget analysis used 1,000 stratified image subsamples at budgets 25, 50, 100, 150, and 200. It reported absolute error relative to the full equivalent-area-side paired contrast, sign recovery, and branch-order recovery. No nested bootstrap was run inside each subsample. A separate pre-access planning analysis reported discovery-bootstrap standard deviations and expected interval half-widths; it was not a power calculation.

3.9. Evidence governance and the attempted prospective audit

After discovery and instrument qualification, the replay implementation, numerical contract, sampling, taxonomy, bootstrap procedures, checkpoint identity, and analysis code were committed in an isolated repository before lockbox selection. A disjoint 300-image outcome-blind AI-TOD-v2 sample was selected with frozen stratum allocation and zero overlap with the discovery images, and one sealed analyzer invocation was permitted. The invocation wrote an irreversible outcome-access receipt, then terminated after outcome access but before producing any estimate or verdict. Because the prospective one-time rule prohibited rerun, no confirmatory lockbox verdict exists. Post-access repair checks qualify the software only and do not restore prospective status; the complete receipt, failure, repair, and exploratory details are reported in the Supplementary Material.

OpenAI Codex was used for code drafting and statistical-workflow assistance; all executed analyses and frozen outputs were reviewed by the authors.

4. Results

Unless stated otherwise, the reported contrasts are descriptive measurements of frozen native assignment states. Image-cluster intervals quantify sampling uncertainty conditional on the audited checkpoint. O2O–O2M contrasts compare states with different supervision semantics, first-divergence labels identify the earliest instrumented change, and cardinal distances follow four grid-resolved rays. These conventions govern the Results below and are repeated only where a local support or conditioning restriction changes the interpretation.

4.1. Qualification of the native replay audit

The qualification suite established that the replay instrument reproduces the native state before scientific interpretation. The analytical suite recovered all six oracle cases and seven property invariants, and it detected all six prespecified mutations. Native execution, production tracing, and the independent NumPy reference produced zero mismatch across 300 base states and 5,251 deterministic directional states. The independent implementation also reproduced

Table 1
Primary AI-TOD-v2 fragility rates and stress-contract contrasts in percentage points. Brackets are 95% stratified image-cluster bootstrap intervals. Gaps are 8–16 minus 16–32 px; positive values denote higher fragility in the 8–16 px group. The paired gap is O2O minus O2M. These are discovery estimates. Support: fixed primary, 6,378 common-valid GTs from 288 contributing images; equivalent-area primary, 6,377 common-valid GTs from 288 contributing images.

Stress contract	Branch	8–16 rate	16–32 rate	Scale gap [95% CI]	Paired gap [95% CI]
Fixed 1 px	O2O	22.14	7.77	14.37 [10.84, 17.78]	12.95 [6.52, 19.16]
Fixed 1 px	O2M	71.74	70.32	1.42 [-3.18, 6.31]	–
Equivalent-area side $\kappa = 0.0625$	O2O	9.11	8.98	0.13 [-3.41, 3.15]	17.58 [11.06, 23.98]
Equivalent-area side $\kappa = 0.0625$	O2M	57.93	75.37	-17.45 [-22.72, -11.71]	–

all 4,408 continuous-boundary qualification trajectories under the then-current schedule. The focal-row engine then matched complete native replay at every state in 3,072 full-grid scenarios, with byte-identical event and curve files. Grid resolution was assessed separately and led to the full-domain 1/1024 primary analysis.

4.2. Stress parameterisation exposes branch-specific response structure

The apparent scale response changed substantially with the stress contract, while the two native branches remained descriptively distinct. On the AI-TOD-v2 seed-0 discovery sample, fixed one-pixel replay produced an O2O 8–16 minus 16–32 px fragility contrast of 14.37 pp. Under equivalent-area-side replay, the O2O contrast was 0.13 pp and its interval included zero. O2M instead changed from an uncertain fixed contrast of 1.42 pp to an equivalent-area-side contrast of -17.45 pp. The paired O2O-minus-O2M contrast was positive under both contracts (Table 1), but fixed stress was dominated by the O2O response whereas equivalent-area-side stress was dominated by the reverse O2M response. The two contracts were not pooled because their displacements answer different questions.

The high absolute O2M assigned-set Top-1 fragility rates in Table 1 are object-level fragility estimates, not directional flip rates, and should not be interpreted as loss of positive supervision. Assigned-set Top-1 is a rank probe inside a multi-positive state; it is not equivalent to loss of positive supervision. Figure 5 separately measures whether the positive set itself persisted.

Direction-normalised replay changed the O2O gap to 2.79 pp [0.44, 4.96], while the O2M gap remained -17.04 pp [-22.16, -11.58] and the paired gap was 19.83 pp [13.65, 25.83]. A separate paired composition sensitivity standardised both size groups to the same aspect-ratio distribution; its paired gaps were 17.53 pp [11.07, 24.08] for equivalent-area-side replay and 19.72 pp [13.67, 25.77] for direction-normalised replay. Rectangular-box parameterisation therefore affected the absolute O2O response, with the largest point difference among elongated objects, but did not remove the paired branch pattern.

Across the three-point equivalent-area-side grid, all six O2M contrasts were negative and all six paired branch contrasts were positive. On AI-TOD-v2, paired contrasts ranged from 17.27 to 18.19 pp; on VisDrone they ranged from 34.79 to 37.25 pp. Absolute O2O contrasts varied with both κ and dataset: the AI-TOD-v2 intervals at $\kappa = 0.03125$ and 0.0625 included zero, whereas the VisDrone contrasts were positive at all three magnitudes. The paired direction was stable over the tested grid, whereas the absolute O2O response was not.

The compatible YOLOv10-S contract showed the same equivalent-area-side branch pattern: O2O was 0.94 pp with an interval spanning zero, O2M was -23.50 pp, and the paired contrast was 24.44 pp. This provides replication across two compatible detector contracts without establishing architecture invariance.

The branch pattern was present at all six physically archived stages spanning the first completed epoch to the final training state. Across those stages, mAP$_{50:95}$ changed from 0.112 to 0.280, while fixed paired contrasts ranged from 11.84 to 15.09 pp and equivalent-area-side paired contrasts from 17.34 to 21.35 pp. No checkpoint was available between epochs 100 and 280, so no continuity or interpolation is claimed across that interval. This six-stage archive is distinct from the retained files named `best.pt`: its final row uses the archived `last.pt` training state labelled 300, whereas the separate seed sensitivity uses post-training-selected artifacts with embedded zero-based epoch 280.

Across every protocol-eligible released seed, the fixed paired contrast on the strict 6,318-GT intersection was 12.47, 12.44, 13.19, and 12.88 pp for seeds 0–3; the equal-seed mean was 12.75 pp [6.27, 18.84]. On the strict 6,317-GT equivalent-area intersection, paired contrasts were 17.20, 17.51, 17.87, and 19.18 pp; the equal-seed mean was 17.94 pp [11.51, 24.23]. Equivalent-area O2M gaps were negative in all four seeds, and the positive-set-turnover contrast remained positive in all four. Both 20,000-replicate crossed seed and image-cluster bootstraps therefore retained the branch direction at the two endpoints and the Rank–Set direction across the eligible checkpoints. These are $S = 4$ sensitivities; eligibility, pathway, and boundary-distance estimates remain conditional on seed 0.

Figure 2 summarises these four sensitivity axes. Panel C uses categorical positions and marks the absent epoch 100–280 archives; no interpolation is implied.

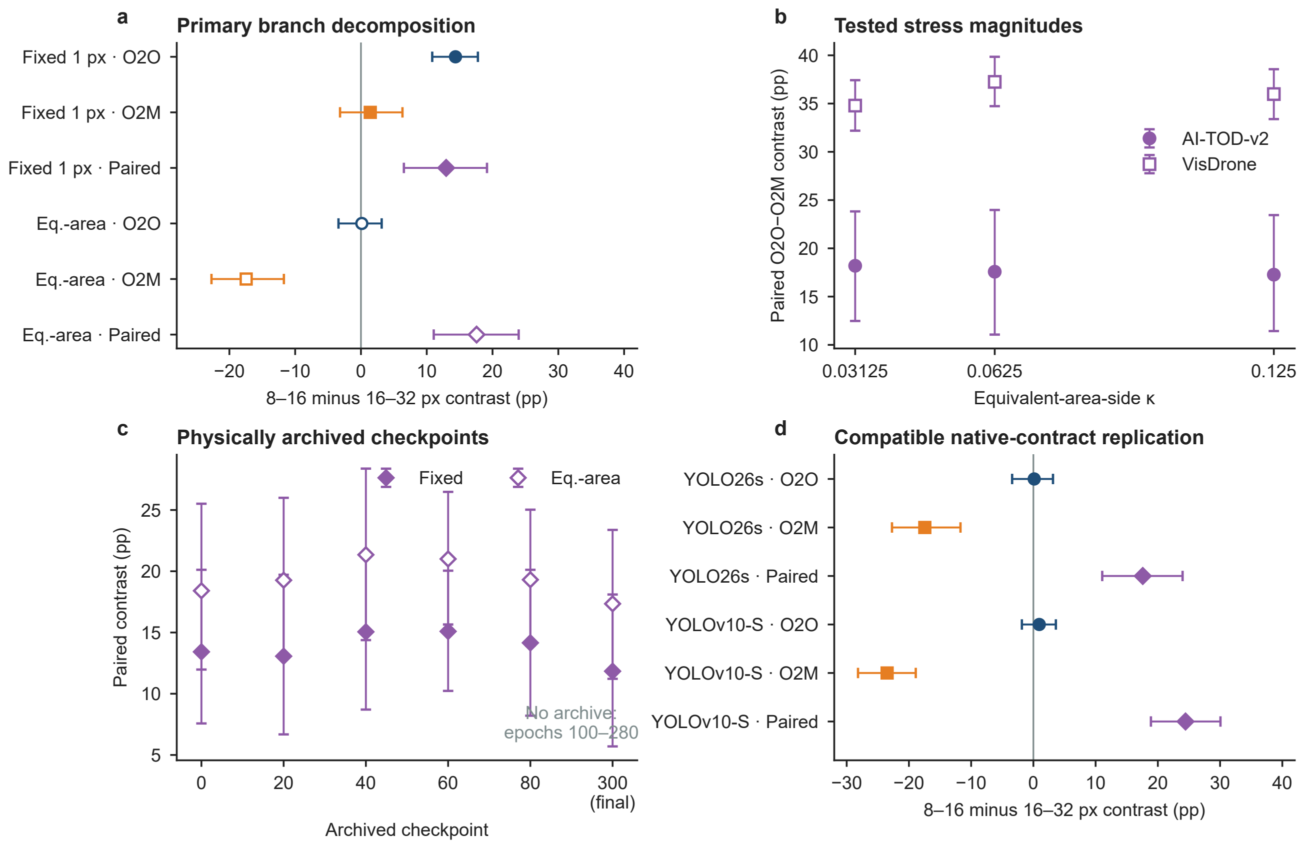

Figure 2: The O2O-minus-O2M contrast remained positive across the audited stress magnitudes, datasets, archived checkpoints, and detector contracts, while branch-wise responses differed between stress contracts. (A) AI-TOD-v2 O2O, O2M, and paired scale contrasts under fixed one-pixel and equivalent-area-side replay. (B) Paired contrasts at the three prespecified equivalent-area-side κ values on AI-TOD-v2 and VisDrone. The values occupy their true positions on a linear numerical axis; no connecting curve or continuous trend model is fitted. (C) Paired contrasts at six equally spaced categorical archived checkpoints labelled 0, 20, 40, 60, 80, and 300 (final); no checkpoints were archived for epochs 100–280. (D) Equivalent-area-side replication under compatible YOLO26s and YOLOv10-S detector contracts. Points are inverse-inclusion-weighted estimates and whiskers are 95% stratified image-cluster bootstrap intervals.

4.3. Eligibility locking and first-observed pathway localisation

Hard eligibility boundaries accounted for much of the fixed-pixel response on the lock-analysis support, while equivalent-area-side replay shifted pathway localisation toward within-set rank changes. The lock analysis used 7,434 fixed-arm objects from 300 selected images, whereas the primary estimate used 6,378 common-valid O2O/O2M objects from 288 contributing images. On the lock support, the native fixed-pixel O2O gap was 17.10 pp and the locked gap was 3.52 pp. Their difference was 13.58 pp, corresponding to 79.44% [70.51%, 91.92%] of the 17.10 pp denominator, not of the separate 14.37 pp primary gap. The equivalent-area-side reduction was 3.15 pp [1.42, 4.83]. The 2.95-pp native equivalent-area value is a property of the independent lock-analysis support and is not a re-estimate of the 0.13-pp common-valid primary contrast.

The observational eligibility-stable subset retained 6,217 of 7,434 fixed-replay objects and 6,999 of 7,432 equivalent-area-side objects. Its fixed O2O scale contrast was 3.36 pp [1.11, 5.48], whereas its equivalent-area-side contrast was 1.05 pp [−1.19, 2.96]. The residual fixed response was smaller than the native response but remained positive. Because eligibility stability is conditioned after replay, these estimates have a different, potentially selected estimand.

First-divergence anatomy showed the corresponding change in pathway composition. Eligibility was the first observed difference for 67.09% [63.02%, 71.07%] of fixed-pixel identity exchanges and 51.01% [46.06%, 55.94%] under equivalent-area-side replay. Within-set geometric rank reversal was first for 30.13% [26.14%, 34.36%] and 45.37% [40.62%, 50.26%], respectively. Because first divergence records the earliest stage in a fixed hierarchy, these shares are order-dependent localisations.

The nonexclusive audit showed substantial downstream co-occurrence. Under fixed replay, a within-set rank change occurred on 75.12% [70.98%, 79.35%] of identity-exchange paths. It was first on only 30.13%, and eligibility co-occurred on 44.63% [41.07%, 48.81%]. Under equivalent-area-side replay, any-stage within-set change was 77.44% [73.15%,

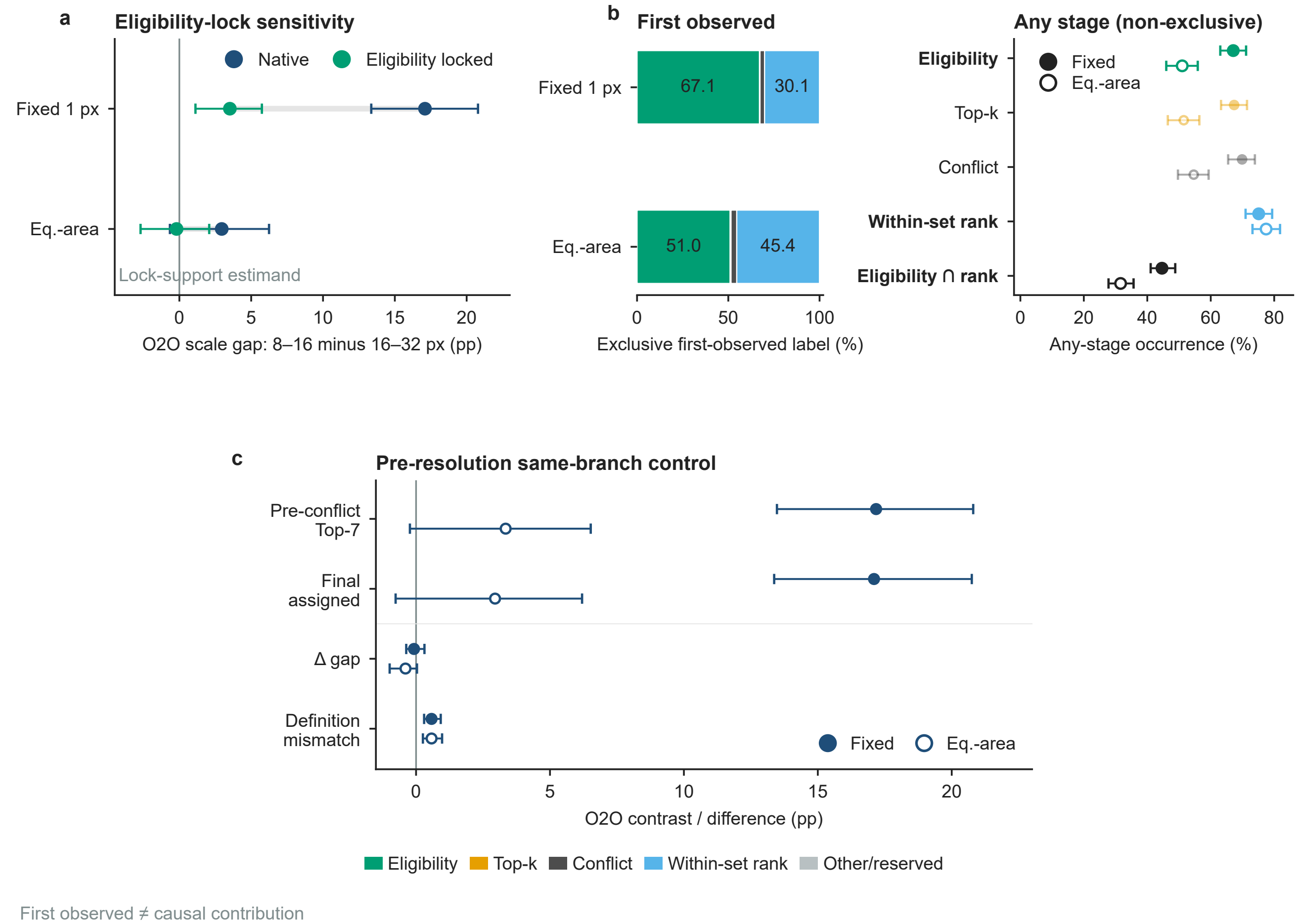

Figure 3: Eligibility locking removed most of the fixed-pixel O2O contrast, while within-set changes occurred more often anywhere along an exchange path than exclusive first labels suggested. (A) Native and eligibility-locked O2O contrasts on the same lock-analysis support. (B) Exclusive first-divergence composition (left) and nonexclusive any-stage occurrence for eligibility, Top-k, conflict, within-set rank, and eligibility–rank co-occurrence (right). (C) Pre-conflict Top-7 winner and final post-conflict assigned contrasts provide a same-branch pre-resolution control. Filled markers denote fixed one-pixel replay and open markers equivalent-area-side replay. Whiskers are 95% stratified image-cluster bootstrap intervals. First divergence localises the earliest observed difference and is not a causal contribution.

81.84%]. It was first on 45.37%, and eligibility co-occurred on 31.59% [27.78%, 35.68%]. The gap between first-label and any-stage occurrence shows how pipeline order changes the localisation summary.

The aggregate O2O branch response was already present before final conflict resolution. The assigned-minus-pre-conflict scale-gap difference was −0.077 pp [−0.367, 0.313] for fixed replay. Under equivalent-area-side replay, it was −0.395 pp [−0.986, 0.034]. Object-level fragility disagreement was 0.578% under both contracts. All 134 fixed directional disagreements and all 96 equivalent-area-side disagreements coincided with the instrumented conflict stage. Conflict localised this small set of disagreements but did not explain the aggregate branch-scale pattern.

These boundary-localisation results are shown in Figure 3. The post-replay eligibility-stable subset remains a distinct selected estimand and is reported numerically in the Supplementary Material rather than repeated in the main figure.

4.4. O2O and O2M occupy different cardinal boundary structures

Assigned O2O identity and O2M assigned-set Top-1 showed different first-departure persistence along the audited cardinal rays. The analysis included 29,724 directional trajectories from 7,431 focal GTs. Through $\tau = 0.125$, O2O had an RMCBD of 0.11985 (95% CI 0.11932–0.12032), versus 0.10025 (0.09864–0.10180) for O2M. The O2O-minus-O2M contrast was 0.01745 (0.01586–0.01898) for 8–16 px objects and 0.03319 (0.02937–0.03652) for 16–32 px objects. By τ, O2O boundaries had been observed for 9.05% (8.28–9.92%) of tiny and 6.39% (4.61–8.49%) of small trajectories; the corresponding O2M rates were 32.75% (31.27–34.22%) and 48.25% (44.89–51.40%). The branches differed in both observed cardinal persistence and

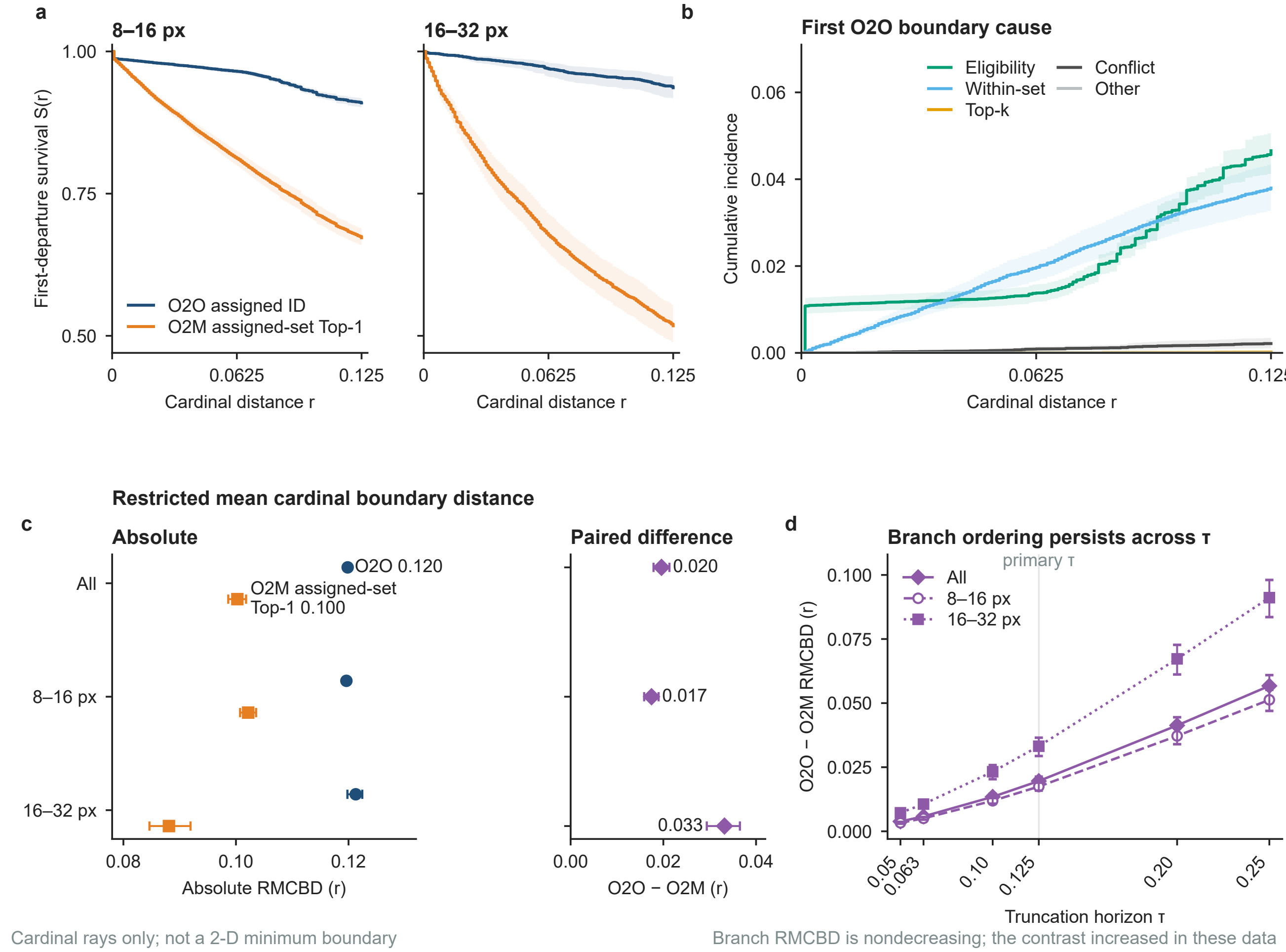

Figure 4: Post-conflict assigned O2O identities persisted farther along cardinal rays than O2M assigned-positive Top-1 identities across the full-domain 1/1024 analysis. (A) Weighted Kaplan–Meier summaries of remaining free of a first observed state departure under the recorded administrative and geometric censoring rules, shown as first-departure survival $S(r)$ for O2O and O2M in the 8–16 and 16–32 px groups. (B) Aalen–Johansen cumulative incidence of the first observed O2O boundary causes. (C) Absolute branch-specific restricted mean cardinal boundary distance (RMCBD) and paired O2O-minus-O2M contrasts through the primary horizon $\tau = 0.125$. (D) Paired contrasts across six prespecified horizons on a linear numerical τ axis. Each branch-specific RMCBD is nondecreasing in τ by construction, whereas monotonicity of their difference is not guaranteed; the paired contrast increased over the observed range. Distances follow four cardinal rays and are not two-dimensional minimum boundaries. KM/AJ/RMCBD are descriptive summaries under the recorded censoring rules.

scale structure. Exact RMCBD values and intervals are tabulated in the Supplementary Material.

Figure 4 summarises full-domain 1/1024 identity survival, cause-specific incidence, restricted mean distance, and truncation-horizon sensitivity.

4.5. O2M Rank–Set turnover and positive-count conditioning

O2M Top-1 turnover and positive-set turnover were distinct native states under both stress contracts. In the canonical 16–32 minus 8–16 px orientation, fixed replay gave Jaccard and base-retention contrasts of +4.19 and +2.26 pp, whereas equivalent-area-side replay gave −5.91 and −4.93 pp. The equivalent-area-side assigned-set Top-1 flip contrast was +12.80 pp, while the rank-only share among flips was −22.50 pp. Smaller objects had fewer Top-1 flips but a larger rank-only fraction among those flips; larger objects combined greater Top-1 turnover with lower set persistence.

The three-stage count analysis separated support restriction from count-composition standardisation. Native positive count averaged 3.07 for 8–16 px and 6.54 for 16–32 px objects. Restricting the object-level Top-1 fragility analysis from full support to the adequate exact-K = 3–7 overlap changed the 16–32-minus-8–16 gap from 17.45 pp [11.85, 22.73] to 4.53 pp [−1.46, 10.52]; standardising K composition within that overlap changed it further to 2.16 pp [−4.74, 9.71]. The direction-level sequence was 11.59 pp [8.42, 14.66], 3.59 pp [0.80, 6.63], and 0.88 pp [−2.18, 4.47]. The overlap covered 60.3% and 64.3% of the two size groups. Positive-set-turnover gaps instead increased from 33.16 pp [31.35, 34.79] on full support to 36.82 pp [34.73,

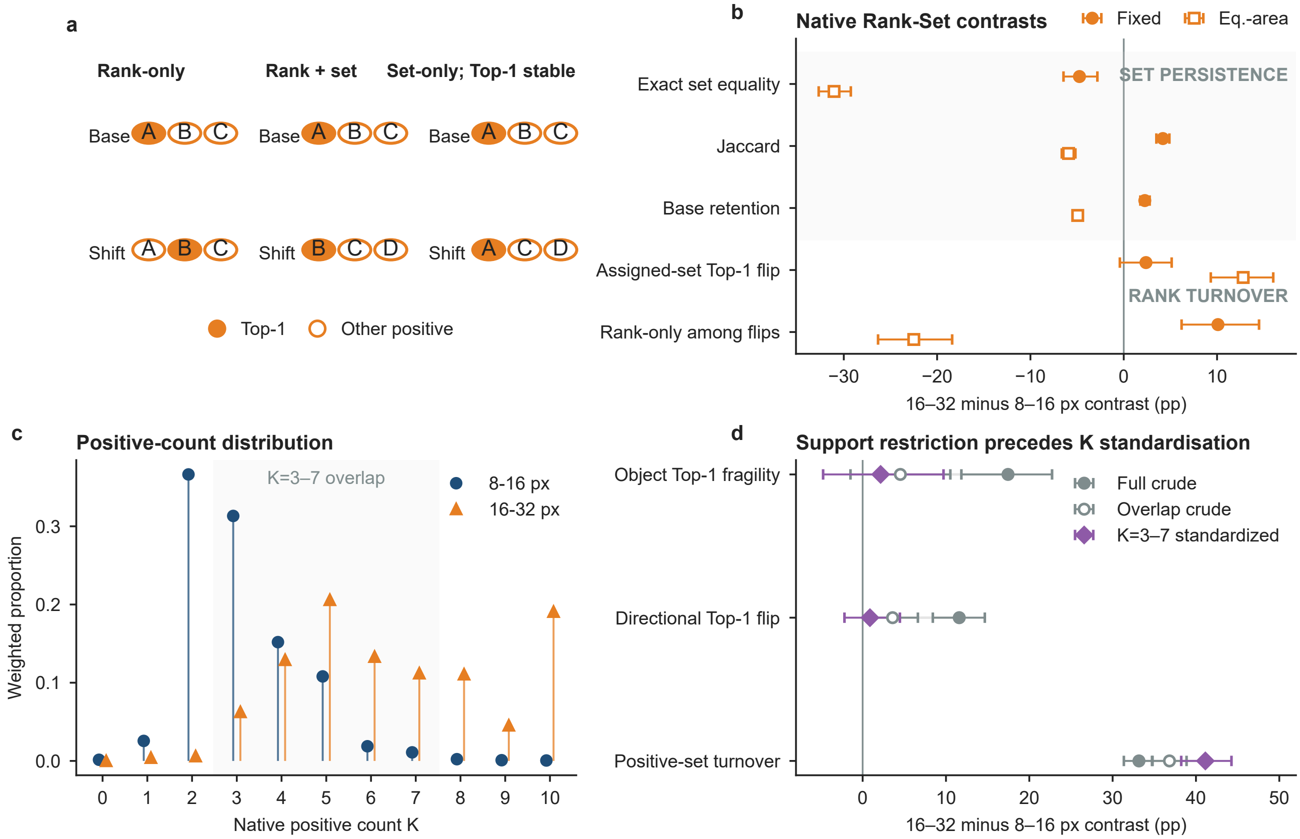

Figure 5: Support restriction accounted for most attenuation of the O2M Top-1 contrast, while positive-set turnover remained large. (A) Schematic distinction among rank-only turnover, simultaneous rank-and-set turnover, and set-only turnover with stable assigned-set Top-1. Filled circles denote Top-1 and open circles other positives. (B) Native O2M Rank–Set contrasts under fixed and equivalent-area-side replay, grouped into set persistence and rank turnover. (C) Weighted native positive-count distributions; $K = 0$ remains visible and shading marks the $K = 3$–7 overlap used for standardisation. (D) Full-support crude, overlap-support crude, and overlap-support count-standardised contrasts. Panels B and D use 16–32 minus 8–16 px throughout. The three markers separate support restriction from within-overlap K standardisation; neither operation identifies a causal count intervention.

38.87] on overlap support and 41.13 pp [38.23, 44.26] after standardisation. Most Top-1 attenuation therefore arose from the support restriction; the additional within-overlap standardisation effect was smaller. Neither change identifies a causal cardinality effect, and the frozen tables lack the complete q ordering required for a genuine fixed-K assigner replay.

Figure 5 separates rank turnover from set persistence and shows how count conditioning affects their scale contrasts.

The strict same-stride analysis retained 3,835 paired GTs. Its O2O scale contrast was +10.23 pp [7.30, 13.32], its O2M contrast was −0.12 pp [−6.83, 5.78], and the paired difference was +10.35 pp [3.54, 18.17]. Cross-stride switching alone therefore did not account for the branch contrast, although post-replay conditioning can induce selection and does not exclude a feature-pyramid-level contribution.

4.6. Native margin aligns with observed rank separation, with a narrower utility boundary

Assigned-relative margin aligned with observed fixed-pair rank distance, but this conditional association did not translate into stable incremental diagnostic value. Only 1,593 of 29,724 trajectories had an observed fixed-pair q-order crossing; 7,439 were competing-censored, 16,548 right-censored, and 4,144 undefined. Within the observed-crossing subset, margin correlated with crossing distance at $\rho = 0.628$ [0.584, 0.669]. Its correlation with observed eligibility-boundary distance was $\rho = 0.150$ [0.111, 0.191], a difference of 0.477 [0.408, 0.543]. These are conditional observed-case associations rather than censoring-adjusted population correlations.

Replacing a linear margin term with a training-fold-fitted restricted cubic spline did not improve held-out metrics consistently. For fragility, spline-minus-linear ROC-AUC and PR-AUC differences were −0.00756 [−0.01411, −0.00138] and −0.00415 [−0.00813, −0.00109]. Coverage-based-miss PR-AUC was lower by −0.00541 [−0.01298, −0.00027]. The log-loss intervals spanned zero for both outcomes. This negative sensitivity limits claims that a more flexible margin shape supplies additional diagnostic utility.

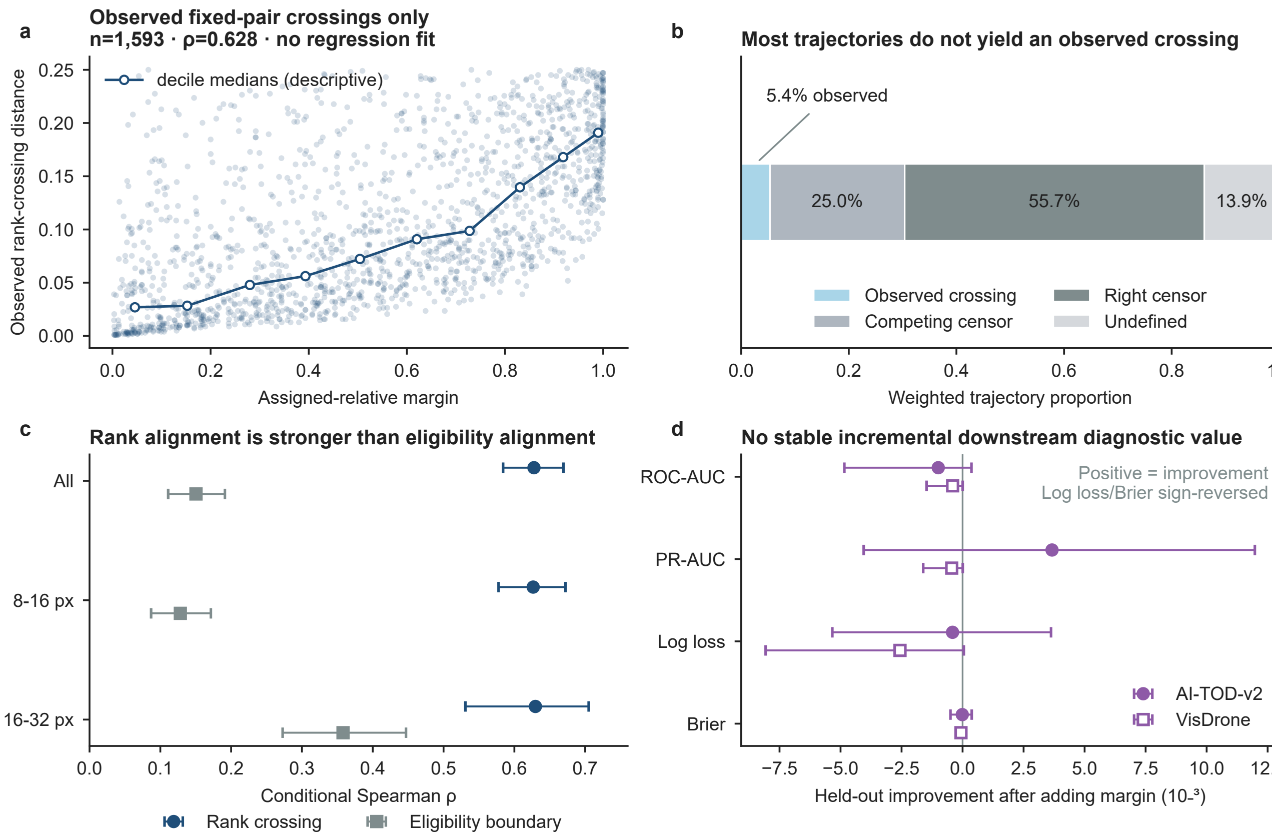

Figure 6: Assigned-relative margin aligned with observed fixed-pair rank-crossing distance but supplied no stable held-out diagnostic improvement. (A) Assigned-relative margin versus observed rank-crossing distance among 1,593 trajectories with an observed crossing. The line joins ten margin-decile medians as a descriptive summary; no regression curve is fitted. (B) Weighted composition of observed crossing, competing censoring, right censoring, and undefined states. (C) Conditional Spearman associations with observed rank crossing and eligibility-boundary distance on their separate observed supports. (D) Held-out diagnostic changes after adding margin on AI-TOD-v2 and VisDrone; log-loss and Brier signs are oriented so rightward values indicate improvement. These analyses concern conditional association and diagnostic increment, not a guaranteed predictor or intervention effect.

Figure 6 separates margin's observed-crossing rank-separation alignment from its more limited incremental diagnostic utility.

4.7. Locality and practical audit scope

The remaining analyses delimit the audit's locality, screening budget, and prospective evidence status. Focal replay changed a non-focal O2O identity in 52 of 3549250 pair-events. The overall weighted pair rate was 0.001465% [0.000824%, 0.002383%]. All observed changes occurred among the 5,679 pairs sharing at least one native pre-conflict candidate claim; their rate was 0.9159% [0.5787%, 1.3248%]. No nonsharing event was observed in this sample.

At a 100-image audit budget, all 1,000 subsamples recovered the sign and O2O-versus-O2M branch ordering of the full equivalent-area-side paired contrast. The median absolute error was 3.43 pp and the 95th-percentile error was 8.41 pp. Reduced budgets can screen a dominant response signature, whereas precise effect estimation still requires the full design.

5. Discussion

5.1. Stress response depends on the measurement contract

The fixed-pixel O2O scale gradient and its attenuation under equivalent-area-side replay measure different perturbation contracts rather than competing estimates of one intrinsic quantity. A one-pixel shift is a larger fractional displacement for a smaller object. Equivalent-area-side replay scales the displacement by $\sqrt{wh}$, although it does not equalise the width- and height-relative fractions of rectangular objects. The fixed response was dominated by O2O, whereas the equivalent-area-side response was dominated by a reverse O2M contrast. The apparent scale effect therefore depended on how geometric stress was defined.

This replay should also be distinguished from training augmentation. Mosaic, scaling, translation, and related transformations jointly change pixels, geometry, features, and predictions. The audit instead freezes these quantities and moves only the focal GT centre. It measures local discontinuities of the resolved assignment function rather than SGD instability or the frequency with which training augmentation crosses an assignment boundary.

5.2. Native assignment stability is state-specific

Eligibility locking and pathway localisation separated two sources of assignment change. On the lock-analysis support, holding eligibility fixed removed most of the fixed-pixel O2O scale gap. Under equivalent-area-side replay, the first observed difference shifted away from eligibility and toward within-set rank reversal. The nonexclusive audit further showed that eligibility and rank changes often occurred on the same path. Operationally, the two analyses answer different questions: the lock tests sensitivity to eligibility state, while first divergence records the earliest instrumented transition.

The full-domain grid extended this hierarchy from endpoint flips to boundary persistence. The post-conflict assigned O2O identity had a larger restricted mean cardinal boundary distance than O2M assigned-set Top-1, and the separation was larger for 16–32 px objects. The ordering persisted across all six truncation points. These results show that the two audited states occupied different observed cardinal geometries over a range of equivalent-area-side distances, rather than differing only at one selected displacement.

For the diagnostic questions considered here, a single endpoint candidate-change variable is insufficient to distinguish the audited native states. Under equivalent-area-side stress, 16–32 px objects showed more Top-1 turnover and lower assigned-set persistence. Restricting both groups to the $K = 3$–7 overlap produced most of the Top-1 attenuation; standardising K composition within that overlap produced a smaller additional change, while positive-set turnover remained large. Operationally, this separates support selection from count composition and both from a change in which candidates remain positive. The pre-resolution control placed conflict in the same state hierarchy: pre-conflict and final assigned gaps were nearly identical, while the small set of disagreements coincided with conflict resolution.

5.3. What the O2O–O2M branch contrast means

O2O and O2M are not exchangeable treatment arms. They use different learned scores, Top-k rules, post-conflict cardinalities, and supervision semantics. Their paired contrast is therefore a descriptive comparison between two native branch states, not an effect of replacing one branch with the other and not a direct ranking of which branch is intrinsically more stable.

The positive paired direction is nevertheless not guaranteed by the definitions. Either branch response can vary with the stress contract, dataset, checkpoint, and audited state. Its repeated direction across the tested stress magnitudes, datasets, archived stages, compatible detector contracts, and protocol-eligible training seeds indicates a systematic difference in scale-by-stress response under those contracts. Dense-grid persistence supports the same distinction beyond a single endpoint. The three-stage positive-count analysis shows that support restriction and cardinality composition both change the O2M Top-1 contrast, so the branch contrast cannot identify which individual branch property produced the observed separation.

5.4. Margin localises rank separation without establishing utility

Assigned-relative margin had a narrow geometric interpretation among the 1,593 trajectories with an observed crossing of the fixed assigned/runner pair. Larger margin was associated with a more distant rank-crossing boundary, whereas its association with eligibility-boundary distance was much weaker. This result concerns observed comparable pairs. Competing censoring, right censoring, and undefined competition prevent it from serving as a population correlation for all trajectories.

The observed-crossing alignment did not supply stable downstream diagnostic improvement. A restricted cubic spline did not improve held-out ROC-AUC or PR-AUC over the linear-margin specification, and adding margin after native score and geometry produced intervals that spanned zero across the tested metrics. Margin can therefore act as a local coordinate for pairwise rank separation without serving as a calibrated failure probability, a general uncertainty score, or evidence for an intervention. When no legal runner-up exists, the comparison is undefined rather than extreme.

5.5. Practical use of a read-only assignment-state audit

The audit provides a concrete workflow for evaluating an assigner or checkpoint without retraining it. A developer can first identify whether a change enters through eligibility, Top-k, within-set rank, positive-set membership, or conflict resolution. The same comparison can then be repeated under fixed-pixel and equivalent-area-side stress to determine whether an apparent scale effect depends on the measurement coordinate. For O2M, Top-1 turnover and positive-set turnover should be inspected separately. Finally, Kaplan–Meier persistence, cause-specific incidence, RMCBD, and pathway composition can show whether a modification moved a boundary or changed the state that fails first. These outputs define diagnostic targets, not intervention prescriptions. An eligibility-dominated change points to the candidate region or eligibility rule; a rank-only change points to score/order sensitivity among retained positives; and positive-set turnover points to supervision-support construction. This workflow turns the broad observation that "assignment changed" into state-specific diagnostic evidence. It does not require an AP claim or prescribe a replacement assigner. Methods that modify similarity, positive construction, density, or supervision schedules may affect different layers even when their output-level behaviour appears similar [1, 2, 3, 4, 9].

Spillover was rare across all focal–non-focal pairs and concentrated among pairs sharing a native pre-conflict candidate claim, which supports a local coupling interpretation in this sample. A 100-image audit consistently recovered the dominant branch direction, but its upper-tail estimation error remained too large for precise effect estimation. Reduced audits may therefore screen checkpoints for a strong signature; scientific comparison still requires the full sampling and uncertainty design.

The principal mechanistic estimates remain discovery results conditional on the audited seed-0 checkpoint. Cross-dataset, archived-stage, detector-contract, and four-seed analyses provide sensitivity or replication; prospective-confirmation status is reported in the Limitations and Supplementary Material.

6. Limitations

The audit is diagnostic and read-only. It measures frozen native assignment states under controlled stress; it does not identify a causal training mechanism, estimate annotation noise, or show that changing an assignment state would improve detection. The O2O–O2M comparison is not a symmetric treatment comparison because the branches use different learned scores, Top-k values, post-conflict cardinalities, and supervision semantics. Exact-$K = 3$–7 restriction and subsequent count standardisation are descriptive operations on different target populations, and the frozen tables do not contain the complete q ordering needed for a genuine fixed-K replay. The paired branch contrast therefore remains descriptive.

Generalisability is bounded by the prespecified population and the available contracts. The principal scale domain is $s_g \in [8, 32)$ px; objects below 8 px lie outside the population estimand. AI-TOD-v2 is the discovery dataset, while VisDrone, three equivalent-area-side magnitudes, six archived YOLO26s stages, one compatible YOLOv10-S contract, and four eligible training seeds provide sensitivity evidence. The archived stages do not cover epochs 100–280. Crossed seed–image bootstraps propagate training-seed variation for the fixed and equivalent-area-side endpoints and Rank–Set summaries, but pathway and cardinal-boundary intervals remain conditional on seed 0. First-divergence proportions should not be assumed invariant across datasets, seeds, or detector contracts.

The stress contracts do not isolate object size from class, stride, aspect ratio, border distance, scene density, or spatial location. Direction-normalised replay and aspect-ratio standardisation address only the known difference between equivalent-area-side displacement and axis-specific fractions for rectangular objects. The paired direction persisted, whereas the absolute O2O gap changed with the stress parameterisation; neither contract is canonical. Eligibility-stable and same-stride analyses condition on observed replay states and may induce selection. Object-level fragility also has an opportunity-count dependence because illegal shifts are omitted rather than clipped; the four-valid-direction sensitivity reduces but does not eliminate this issue. First-divergence and conflict-stage labels localise observed transitions and are not additive causal contributions. The pre-resolution control identifies where a small set of disagreements occurs, not a causal conflict-resolver effect. Spillover was rare and concentrated among pairs sharing a native pre-conflict claim; zero events among nonsharing pairs do not establish independence or equivalence. Reduced-budget analyses support screening of the branch ordering but do not define a universal sample-size rule.

Cardinal persistence is measured on four rays through $r_{\max} = 0.25$, not as the minimum distance in a two-dimensional perturbation plane. The exhaustive $1/1024$ grid cannot skip an event visible at that resolution, but it cannot resolve a transient $A \to B \to A$ interval narrower than one grid step. Trajectories without an observed event are right-censored, and competing first causes are retained. The legal-radius censoring point can depend on image position. Accordingly, KM/AJ/RMCBD are descriptive summaries under the recorded administrative and geometric censoring rules, not censoring-independent population persistence estimates; raw first-event incidence, cause counts, and return transitions remain separate reports. The first-departure label is the earliest observed stage in a fixed hierarchy, so it is a localisation summary rather than a decomposition of total instability. The primary restricted mean uses $\tau = 0.125$; truncation-point sensitivities describe estimator dependence but do not remove directional, censoring, or finite-grid limits.

Assigned-relative margin–ρ_R association is conditional on the 1,593 observed fixed-pair crossings and is not a censoring-adjusted population correlation. The coverage-based-miss outcome is not the standard one-to-one evaluator false negative, and the spline sensitivity compares only two prespecified margin shapes. It therefore does not establish construct validity, calibrated uncertainty, or intervention utility. Exact reproduction requires the identified checkpoints and separately obtained benchmark images; the public release excludes third-party images, annotations, and raw prediction dumps. The four released checkpoints are bound to the common contract, but checkpoint-selection variability remains outside the image-cluster intervals. Finally, the disjoint prospective audit produced no normative verdict, was not rerun after its single invocation terminated before an estimate or decision, and its post-access repair checks do not restore confirmatory status.

7. Conclusion

We developed a read-only framework for measuring native dense-detector assignment states under controlled stress. The results show that assignment stability is both state specific and stress-contract dependent. Fixed-pixel O2O response was strongly coupled to eligibility on the lock-analysis support, whereas equivalent-area-side replay shifted first-observed localisation toward within-set rank reversal. Full-domain $1/1024$ analysis showed different cardinal persistence structures for the post-conflict assigned O2O identity and O2M assigned-set Top-1. Rank–Set decomposition further separated O2M rank turnover from positive-set persistence and showed that support restriction and positive-count composition affected those summaries differently.

The framework turns an aggregate assignment change into an audit of eligibility, rank, set membership, conflict, and boundary persistence. Assigned-relative margin was associated with observed pairwise rank-separation distance but provided no stable incremental diagnostic value. The primary mechanistic claims remain discovery evidence. Analyses across datasets, archived stages, detector contracts, and four protocol-eligible training seeds provide sensitivity or replication. Broader seed, architecture, and non-cardinal perturbation coverage would be required for stronger generalisation.

CRediT authorship contribution statement

Yede Liu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing–original draft. **Fang Wang:** Validation, Investigation, Writing–review and editing. **Jun Li:** Supervision, Project administration, Writing–review and editing.

Declarations

Funding. This research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability. Code, derived data, figure-source tables, the complete (1/1024) dense-grid records, and four protocol-matched checkpoints corresponding to this article are available from the article-matched GitHub release v3.2.0. The corresponding article source package is archived in the Zenodo record 10.5281/zenodo.22478277. The released code and derived materials are provided under AGPL-3.0-only. Raw AI-TOD-v2 and VisDrone images and annotations are not redistributed and must be obtained from their original providers and providers.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work, the authors used OpenAI Codex (OpenAI; accessed August–September 2026) for advisory assistance with manuscript wording, reproducible code, statistical-workflow implementation, and figure layout. After using this tool, the authors independently reviewed and edited all outputs as needed and take full responsibility for the content of the published article.

Acknowledgements

The authors thank Sichuan Normal University for providing the research environment and computational support for this study.

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